(2012).

2. Shankar, R., Industrial Engineering and Management, 2nd Edition, Galgotia Publications (2016).

Reference Books

2. Grant, E.L., Statistical Quality Control, 3rd Edition, McGraw Hill (2014).

3. Sanders, M. and McCormick, E., Human factors in Engineering, 4th Edition, McGraw Hill (1993).

Evaluation Scheme

S. No	**Assessment**	**Weightage (%)**
1.	MST	30
2.	EST	45
3.	Sessional (tutorial assignments/quizzes)	25

UME723 : Fluid Machines

L	T	P	Cr
3	1	2*	4.0

Course Objective: Students will be exposed to the fundamentals of momentum equation, working principles of the hydropower plant and its components, centrifugal pumps, design parameters of the centrifugal and reciprocating pumps. Course provides basic introduction of subsonic and supersonic flows through nozzles of gases and vapour, the students will have exposure to steam turbine types and their designs, governing, gas turbine cycles and their thermal refinements, jet propulsion. Students will be introduced to centrifugal and axial flow compressors, performance and design characteristics of compressors.

Syllabus

Principles of fluid machines: Impulse momentum equation, Impact of jets.

Hydraulic Turbines: Classification, head losses, efficiencies, hydropower plant, various elements, impulse and reaction turbines, components, selection of design parameters, size calculations, work, efficiency, Similarity relations and unit quantities, specific speed, cavitation.

Hydraulic Pumps: classification, selection, installation, centrifugal pumps, head, vane shape, pressure rise, velocity vector diagrams, work, efficiency, design parameters, multistaging, operation in series and parallel, submersible pumps, NPSH, specific speed.

Reciprocating Pumps: indicator diagram, work, efficiency, effect of acceleration and friction, air vessels.

Compressible Flow: Stagnation Properties, Speed of sound and Mach Number, one dimensional isentropic flow, isentropic flow through nozzles, shock waves, air and steam flow through nozzles, supersaturated flow.

Steam Turbines: Steam nozzles, isentropic flow, critical pressure ratio, maximum discharge, throat and exit areas, effect of friction, supersaturated flow. Stream Turbines, types, impulse turbine, velocity and pressure compounding, reaction turbine, degree of reaction, losses, partial admission factor, overall efficiency, governing.

Compressors and blowers: Centrifugal and axial flow type; characteristic curves of compressors.

Gas Turbines: Brayton cycles and its modifications, jet propulsion, turbo jet, ram jet, turbo-prop.

Course Learning Objectives (CLO)

The students will be able to:

1. Derive and apply thermodynamic and fluid terminology to fluid machines.
2. Determine the parameters affecting performance pumps and turbine.
3. Draw the velocity triangles in turbo machinery stages operating at design and off- design conditions.
4. Determine methods to analyze flow behavior depending upon nature of working fluid and geometric configuration of fluid machinery.

Text Books